Process Rewards for Mathematical Reasoning: GRPO Baselines and Failure Analysis

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Abstract

We compare outcome-only GRPO with process reward mixing for Qwen2.5-Math-1.5B. At about 6M generated tokens, outcome-only training reaches 54.00% on MATH500 and 67.02% on GSM8K; adding a frozen 7B PRM gives 53.80% and 64.97%. Neither difference is conclusive. Saved signals show PRM errors and normalization that gives low-scored wrong responses positive advantages. A bounded, correct-answer-only correction improves accuracy in two follow-up seeds, but the second has smaller, inconclusive gains. We include a matched-scorer control and distinguish signal constraints from evidence of reasoning quality.

1 Research question and motivation

Final-answer verification is cheap, but cannot distinguish valid reasoning from a lucky answer. Process supervision could help, yet an imperfect score can produce a misleading update. We ask whether PRM scores remain useful after GRPO transforms them into advantages, and whether the resulting policy solves more problems correctly.

Our original proposal compared PRM adaptation using successful and failed responses with adaptation using failures alone. We report that pilot alongside the policy experiments. Our contribution is a controlled reward comparison, a training-signal audit, and a bounded advantage variant. Higher answer accuracy alone does not establish more valid reasoning.

2 Related work and scope

GRPO learns from groups of responses without a learned critic (Shao et al., 2024); LoRA reduces the trainable parameter count (Hu et al., 2021). Human process supervision (Lightman et al., 2023) and Qwen's PRM study (Zhang et al., 2025) motivate step feedback, but also leave label reliability and distribution shift as concerns. ProcessBench tests first-error identification rather than only final-answer selection (Zheng et al., 2024). PURE assigns min-form step credit (Cheng et al., 2025); simply taking a minimum score and applying response-level GRPO, as in our B baseline, does not reproduce PURE.

The closest advantage methods are PAPO, which separately normalizes process signals within correct responses (Tan et al., 2026), and PROGRS, which centers process rewards within incorrect responses while retaining correct-response scores (Rezaei et al., 2026). Our correction instead centers raw PRM scores only within correct responses and bounds its magnitude without a process standard deviation. PRPO also aligns process and outcome signals, at finer granularity (Ding et al., 2026). These precedents limit our novelty claim to an empirical comparison and a bounded variant. Dr. GRPO identifies length bias in response-normalized losses (Liu et al., 2025); we retain that loss throughout the reported comparison, so our correction does not resolve this separate issue.

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3 Rewards and policy updates

3.1 Outcome verification and process scoring

For a problem q, the policy samples $G = 8$ normal chain-of-thought responses. Let $v_i = 1$ when response i ends normally and its answer is parseable under the fixed protocol. The binary outcome reward y_i is one if its final boxed answer matches the reference. Truncated or unparseable responses receive zero. For eligible responses, we split steps at canonical double line breaks, append Qwen's step marker, and read the positive-class probabilities:

$$\widetilde{m}_i = \begin{cases} \min_{1 \le k \le K_i} p_\phi(\text{valid} \mid q, s_{i,\le k}), & v_i = 1, \\ 0, & v_i = 0. \end{cases} \qquad R_i^A = y_i, \quad R_i^B = y_i + 0.2\widetilde{m}_i. \tag{1}$$

The zero for an ineligible response is a program placeholder, not a PRM judgment. B uses the unadapted Qwen2.5-Math-PRM-7B. C-mix uses the same reward formula with the adapted C scorer described in Section 6. Each scorer stays frozen during policy training.

3.2 GRPO objective

For each group, we compute one detached advantage per response:

$$\bar{R} = \frac{1}{G}\sum_j R_j, \qquad s_R = \sqrt{\frac{1}{G-1}\sum_j (R_j - \bar{R})^2}, \qquad \widehat{A}_i = \frac{R_i - \bar{R}}{s_R + \delta}, \quad \delta = 10^{-4}. \tag{2}$$

Let $h_{i,t} = (q, o_{i,<t})$, $\rho_{i,t} = \pi_\theta(o_{i,t} \mid h_{i,t}) / \pi_{\text{old}}(o_{i,t} \mid h_{i,t})$, and $u_{i,t} = \pi_{\text{ref}}(o_{i,t} \mid h_{i,t}) / \pi_\theta(o_{i,t} \mid h_{i,t})$. The old-policy denominator is detached, and the reference policy is fixed. With T_i active completion tokens, the selected `grpo` loss is

$$\begin{aligned} \mathcal{L} &= -\frac{1}{G}\sum_i \frac{1}{T_i}\sum_t \left[\ell_{i,t} - \beta\big(u_{i,t} - \log u_{i,t} - 1\big)\right], \\ \ell_{i,t} &= \min\Big\{\rho_{i,t}\,\text{sg}(\widehat{A}_i), \text{clip}(\rho_{i,t}, 1-\epsilon, 1+\epsilon)\,\text{sg}(\widehat{A}_i)\Big\}. \end{aligned} \tag{3}$$

where $\epsilon = 0.2$, $\beta = 0.04$, and sg stops gradients (Shao et al., 2024; Hugging Face, n.d.). The reward need not be differentiable: a positive advantage increases the sampled tokens' log probabilities, while a negative one decreases them. Only policy LoRA parameters are updated. We use one optimizer iteration per rollout batch and no second advantage normalization.

3.3 Bounded process correction

For C-bounded, we first compute outcome advantages and then add a correction. Let $I_+ = \{j : y_j = 1\}$ and let $\bar{m}_+$ be the mean score over I_+. Define

$$\widehat{A}_i^{\text{CB}} = \frac{y_i - \bar{y}}{s_y + \delta} + c_i, \qquad c_i = 0.2\, y_i(\widetilde{m}_i - \bar{m}_+). \tag{4}$$

Set $c_i = 0$ when I_+ is empty, with no normalization after addition. Since scores lie in $[0, 1]$, $|c_i| \le 0.2$ and $\sum_i c_i = 0$. Wrong answers cannot receive positive reward advantages; all-wrong groups have zero reward advantages, although KL may still contribute a gradient. These are algebraic properties, not a guarantee of better mathematics. The rule discards potentially useful process feedback on failed responses and changes gating, scale, and normalization together.

Common training settings. We use the base Qwen2.5-Math-1.5B (Yang et al., 2024), rank-16 LoRA (alpha 32, dropout 0), AdamW, constant learning rate 5×10^{-6}, and gradient clipping at 1. Each update uses four prompts and eight responses per prompt (microbatch 1, accumulation 32). Sampling uses temperature 1, top-p 1, and no top-k restriction. Prompt and completion limits are each 2,048 tokens. The implementation uses TRL 0.23.1, vLLM 0.10.2, and one A100 80 GB. Sampling importance correction is disabled; reward replay does not establish numerical identity between sampling and learning.

4 Experiments and accuracy

We train on 6,993 MATH problems, hold out 500 for validation, and test on MATH500 (500) and GSM8K (1,319) (Hendrycks et al., 2021; Cobbe et al., 2021). Evaluation uses one greedy response, a 2,048-token cap, and the fixed verifier. GSM8K is evaluation-only. All intervals use 10,000 paired question-bootstrap samples (seed 285), measuring question-level rather than training-seed uncertainty.

4.1 Direct PRM mixing: historical V0 comparison

These runs use corrected percentage/unit verification and CRLF handling for PRM inputs; earlier buggy runs are excluded. A/B share initialization and 1,120 question groups, stopping after a complete update reaches 6M tokens. A uses 6,022,435 tokens and 284 updates; B uses 6,008,457 and 280. A has 16 extra groups: generation budgets are approximately matched, updates and total compute are not.

Table 1: V0, seed 17: outcome-only A versus unadapted-PRM B. Accuracy is in percent; differences and intervals are percentage points (pp).

Test	A	B	B−A	Paired 95% interval
MATH500	54.00	53.80	−0.20	[−4.00, +3.60]
GSM8K	67.02	64.97	−2.05	[−4.25, +0.08]

Both intervals include zero, establishing neither benefit nor equivalence. Checks regraded all 3,638 predictions. For 7,304 scored B responses, step spans and text hashes match PRM inputs; replayed advantages match loss inputs without duplicate normalization.

4.2 Adapted scorer and advantage construction: V1 comparison

V1 fixes 288 updates and question order within each seed. C-mix and C-bounded share the adapted scorer, isolating advantage construction. The initial model below uses V1 evaluation; it is not a matched V0 baseline.

Table 2: V1 accuracy (%). Tokens include failures; wrong-positive counts refer to training advantages. C-mix has only seed 17.

Seed	Policy	Tokens (M)	MATH500	GSM8K	Wrong-positive
—	Initial	—	49.00	56.86	—
17	A	6.155	53.00	66.87	0
17	C-mix	6.114	54.00	67.70	526
17	C-bounded	6.167	57.80	69.37	0
42	A	6.309	52.80	67.48	0
42	C-bounded	6.229	54.80	67.93	0

Table 3: V1 paired changes (pp, 95% intervals). Seed 42 is inconclusive; seeds are not pooled as independent test questions.

Seed	Comparison	MATH500	GSM8K
17	C-mix − A	+1.00 [−2.20, 4.20]	+0.83 [−0.91, 2.58]
17	C-bounded − A	+4.80 [1.40, 8.40]	+2.50 [0.45, 4.55]
17	C-bounded − C-mix	+3.80 [0.40, 7.40]	+1.67 [−0.30, 3.64]
42	C-bounded − A	+2.00 [−1.00, 5.00]	+0.45 [−1.52, 2.43]

Why there are two A results. V0 and V1 are independent runs with different stopping rules. V1 also enables prefix caching, chunked prefill, and asynchronous scheduling, with different vLLM memory allocation. Seed-17 follow-up restart histories differ between arms; seed 42 has matched restart boundaries. The two A scores (54.00%/53.00%) therefore do not isolate seed noise or one configuration change. We compare methods within protocols.

5 What goes wrong with the process signal?

5.1 A small reward weight need not produce a small advantage

When a group is all-correct or all-wrong, its outcome reward is constant. Substituting Eq. (1) into Eq. (2) gives

$$\widehat{A}_i^B = \frac{\widetilde{m}_i - \overline{\widetilde{m}}}{s_{\widetilde{m}} + \delta/0.2} = \frac{\widetilde{m}_i - \overline{\widetilde{m}}}{s_{\widetilde{m}} + 5 \times 10^{-4}}. \tag{5}$$

The 0.2 factor mostly cancels. A has zero reward advantages in such groups; B still ranks responses by the PRM. In V0 B, 242 of 1,120 groups (21.61%) have homogeneous outcomes. They contribute 19.92% of the absolute advantage sum, which is not a gradient-norm share. Of 577 wrong responses with positive advantages, 571 come from all-wrong groups and six from mixed groups.

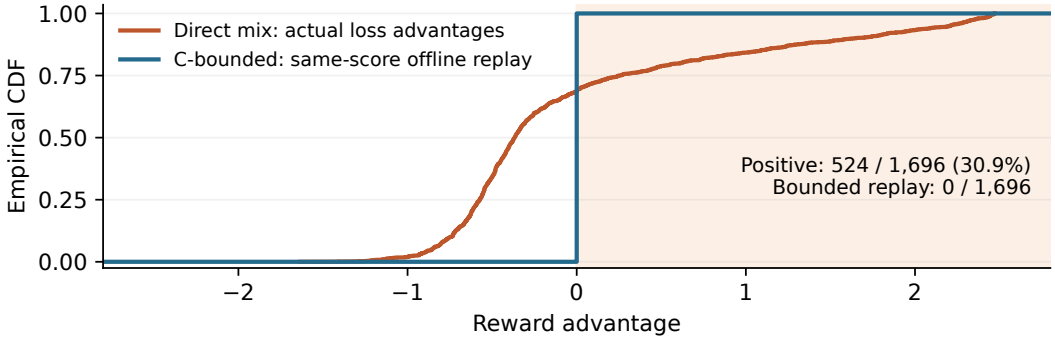


Figure 1: Matched-sample offline replay on the 1,696 responses in C-mix's 212 all-wrong groups (seed 17). We hold responses and adapted PRM scores fixed. Direct mixing gives 524 wrong responses positive advantages; C-bounded gives zero throughout. The bounded curve is a formula replay, not a separately trained policy's distribution or evidence of improved accuracy.

5.2 Score errors versus misleading updates

Positive advantages for wrong final answers are not inherently harmful: a failed response can contain useful reasoning. We therefore inspect concrete, checkable errors rather than interpreting the counts alone. One saved response reports $4 \cdot 6 + 8 \cdot 3 - 28/2 = 36$ and claims its code returns 36; direct evaluation gives 34. Its PRM minimum is 0.9797, but its actual advantage is -1.072. Here the PRM misses an error, yet the update still penalizes the response.

Another response to a nine-vertex coloring problem uses node index 9, outside the valid range 0–8, and claims inconsistent counts of 132 and 12. Exhaustive enumeration of the original graph gives 54. Its minimum PRM score is only 0.0538, yet it receives advantage $+1.070$ in an all-wrong group. This shows how relative scaling can reward a clearly defective response without a high absolute score. These examples establish possible failure mechanisms, not their population frequency or causal contribution to the accuracy gap.

5.3 Formatting is a separate source of error

On GSM8K in V0, B has fewer terminated-but-unparseable outputs than A (53 versus 89), but more parsed wrong answers (368 versus 314) and truncations (41 versus 32). Missing answers alone do not explain the lower B score. Average step counts are similar (11.917/11.899), and PRM scores do not rise monotonically; these observations do not establish reward hacking.

A separate rule-based MATH500 audit finds leading placeholder boxes in 8.2% of initial-model outputs, 30.0% of V1 A, and 63.2% of C-bounded (seed 17). A leading placeholder is an empty/template box within the first 20 non-whitespace characters, followed by at least 100 characters of body text. This is a format shift, not evidence that the model chose an answer before reasoning. In 200 selected correct C-bounded outputs, deleting the placeholder slightly *increases* the mean minimum PRM score: original minus edited is -0.00445 (paired 95% interval $[-0.00890, -0.00009]$). That check does not support the simple explanation that the PRM rewards adding the placeholder.

6 PRM adaptation from the original proposal

We carried out a paired 7B PRM pilot using 512 questions with four policy responses each, plus 2,048 shared step-label anchors (1,024 positive and 1,024 negative). C uses successful and failed responses; D uses failures alone. Both start from the same PRM, freeze the base and classification head, and train rank-8 LoRA (alpha 16) for 512 updates with the same batches, optimizer, and learning rate 2×10^{-5}.

Following LCA's pooling (Jia et al., 2026), let $e_{ik} = 1 - p_{ik}$ and $E_i = \sum_k w_{ik} e_{ik}$, where $w_{ik} = \mathrm{sg}[\mathrm{softmax}_k(e_i/0.2)]$. Success and failure losses are $-\log(1 - E_i)$ and $-\log E_i$, with numerical clipping. C includes both outcome-conditioned terms and balanced step-anchor cross-entropy; D omits only the success term. This is an LCA-inspired adaptation pilot, not a full reproduction; C/D name reward models, distinct from policy C-mix and C-bounded.

Table 4: Fixed ProcessBench diagnostic: 64 valid and 128 invalid solutions. Acceptance means all step scores are ≥ 0.5; first-error accuracy requires the first score below 0.5 to match the label. H is their harmonic mean. This selected subset is not a prevalence estimate.

Scorer	Valid mean min	Valid acceptance	First-error accuracy	H
Unadapted B	0.879	90.63%	67.97%	77.68%
C	0.842	87.50%	73.44%	79.85%
D	0.155	14.06%	35.16%	20.09%

D lowers scores even on human-labeled valid derivations. C preserves much more valid-solution acceptance, but direct mixing still produces wrong-positive advantages (Table 2). D has no full policy-RL run. This single-seed diagnostic retains inherited PRM training exposure; it is not a clean held-out comparison of adaptation algorithms.

7 Costs, limitations, and remaining experiments

Resource accounting. In V0, A takes 2.70 hours and B 2.22 hours; peak allocated memory is 31.60 and 44.88 GiB, respectively. B scores another 5.28M PRM input tokens (7.72M padded), taking 593 seconds including preprocessing and transfer. Shared load prevents a speed comparison. Generation budgets exclude scoring, padding, and backward compute; these costs need separate reporting.

What the results support. V1 RL improves initial-model accuracy. Direct mixing shows no clear benefit; bounded gains are seed-sensitive. C-mix versus C-bounded controls the scorer, but combines gating, magnitude, and normalization changes. Two seeds cannot establish seed-level reliability. The verifier can reject equivalent answers, and accuracy does not establish process validity. Test inspection, inherited PRM exposure, and validation overlap limit generalization claims; further design must use validation only.

Tentative milestones. By November 16, freeze evaluation code and prepare a blinded, independently checked sample of 100 derivations, stratified by method and correctness, reporting agreement and sampling weights. By November 23, subject to the shared-card budget, compare C-bounded with two seed-17 controls: shuffled within-correct PRM rankings and a centered random correction with matched magnitude. Each continues the same checkpoint for 48 updates, with matched question order and validation protocol. By November 30, repeat the more informative control with seed 42 if feasible, then finish error analysis, accuracy intervals, and separate generation/scoring cost totals. These are project targets, not course deadlines.

If real PRM rankings outperform the controls and improve independently assessed process validity without an accuracy loss, that would support a useful process signal. If controls match the gain, the likely benefit is generic signal regularization rather than PRM ranking. If accuracy or validity deteriorates, we will report that limitation rather than claim a successful repair. Code, configurations, and aggregate evidence are available in the project repository.[1]

[1] `https://github.com/mafeifei798/ece285-math-rl`

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